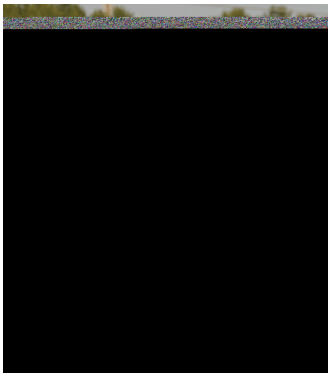
FarmGuard: Unmanned Aerial Robotic Defender for Deer Detection and Deterrence

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The problem

When discussing the challenges posed by deer, several of the farmers we spoke to said the same thing: deer are damaging and eating their crops before they can harvest them. Existing solutions are not universal, and come with drawbacks and tradeoffs. Not all farmers own the land, so solutions like fences are not always possible.



FarmGuard

FarmGuard is a drone-based deer deterrence system that uses cameras and periodic, energy-efficient flight patrols designed to locate deer using computer vision. Once deer are identified, drones switch to herding mode where they set their location and use light and sound stimuli to safely deter the deer away from the farm.

What's next ?

1. Improve the system and test it with deer.
2. Commercial potential for other applications in management of wildlife (birds, boar, etc) that impact agriculture.

Technical Overview

- The path-planning system addresses obstacle avoidance, multi-drone task partitioning, and energy efficiency.
- Each drone is assigned to separate waypoint regions and the UAVs fly at different altitudes to prevent collisions.
- To enable high-fidelity behaviors of deer, we place deer in the simulation and make them to react when the drone gets close We have not yet been able to test the system with real deer.
- After the drones have finished patrolling or herding, or are low on battery, they return to their covered charging stations where they land using onboard cameras and integrated RTK GPS. This keeps them protected from bad weather and ready for deployment.

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